ROMAN ABAGHYAN

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LinkedIn | GitHub

SKILLS

- Ollama
- LLM Integration
- Embeddings & Semantic Search
- Pgvector & Hybrid Search
- Codex CLI, Claude Code
- Agent Configuration & Skills
- MCP Integrations
- System Design
- Python
- Django
- FastAPI
- REST APIs
- SQLAlchemy
- PostgreSQL
- Docker
- Linux
- CI/CD
- AWS

LANGUAGES

Armenian - Native

English - Upper-intermediate

Russian - Upper-intermediate

Education

Yerevan State University

BSc, Informatics and Applied Mathematics
09/2022 - Present

Picsart Academy

Certificate, Computer Science and Programming Fundamentals
04/2023 - 05/2024

PROFILE

AI Software Engineer with a strong Python backend background and hands-on experience building LLM-powered applications, semantic retrieval systems, and AI-assisted development workflows. Experienced in integrating language models, developing reliable APIs, and configuring Codex and Claude Code engineering environments. Focused on agentic software engineering, automation, system architecture, and reliable AI applications.

EXPERIENCE

AI Software Engineer

Ministry of Defence of the Republic of Armenia | Jan 2025 - Present

AI-Powered News Intelligence & Analytics Platform

- Integrated Ollama for LLM-based multi-label news classification and entity extraction.
- Developed structured LLM inference workflows with output validation, retries, and model fallback.
- Implemented embeddings and PostgreSQL/pgvector hybrid retrieval for topic discovery and content analysis.
- Built Django REST APIs with owner-scoped authorization and automated ingestion from web, news, and social sources.
- Developed PostgreSQL search and optimized ORM queries, supported by query-count regression tests.

PROJECTS

Traffic Simulation Lab — AI-Assisted Simulation Platform (In Development)

- Developing a full-stack traffic simulation platform with interactive 2D/3D maps, vehicle movement, traffic signals, and configurable scenarios.
- Configured Codex and Claude Code with specialized engineering agents, reusable skills, and project-specific development instructions.
- Using AI coding agents for implementation, debugging, code review, and verification across backend, frontend, and simulation components.

Personal Finance Platform

- Built a full-stack personal finance platform supporting multi-currency accounts, transactions, budgeting, loans, and financial analytics.
- Implemented an auditable financial ledger, automated recurring transactions, cash-flow forecasting, and financial reporting.
- Developed backend, frontend, and cross-platform mobile functionality using AI-assisted development workflows.

Event Planner Backend

- Developed REST APIs using FastAPI and PostgreSQL to support event creation, scheduling, and management.
- Implemented event scheduling and conflict-resolution logic to identify and handle overlapping events.
- Built asynchronous backend workflows to support event scheduling and processing.
- Containerized the application using Docker to simplify deployment and environment configuration.